Fiscal Year:	FY 2016	**Task Last Updated:**	FY 03/10/2016
PI Name:	Trexler, Matthew D. Ph.D.		
Project Title:	Developing a NASA/NSBRI Integrated Garment Solution to Mitigate Visual Impairment Intracranial Pressure Issues During Long Duration Space Flight		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Cardiovascular Alterations Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	mtrexler@underarmour.com	**Fax:**	FY
PI Organization Type:	INDUSTRY	**Phone:**	410-468-2512
Organization Name:	Under Armour		
PI Address 1:	1020 Hull Street		
PI Address 2:			
PI Web Page:			
City:	Baltimore	**State:**	MD
Zip Code:	21230	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	11/01/2014	**End Date:**	10/31/2015
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC 9-58-CA00004		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Efforts of this project are focused on the design and construction of a light weight, fabric based wearable vacuum system to reduce the intracranial pressure (ICP) of astronauts during space flight, which has been linked to visual impairment both during and after prolonged duration in space. This impairment poses a great danger to the health and safety of astronauts who are spending longer time in microgravity environments. Levine et al. has shown that on Earth intracranial pressure can be lowered in bed ridden patients by simply elevating the head while sleeping. It is estimated that a relative low vacuum (15 mmHg) environment imposed on the lower body during space flight would be enough to reduce ICP to below high risk levels for visual impairment intracranial pressure (VIIP) (i.e., equivalent to the upright posture on Earth). This is due to the fact that the compliance of the intracranial compartment is so low that minimal reductions in the volume dramatically reduce the pressure.		

Task Description:	As of now, orthostatic intolerance is much less of a concern than the ability to gently reduce intracranial pressure on a regular basis (while sitting quietly, or while sleeping) to avoid the VIIP problem. A key challenge has been the ability for soft systems and even some rigid tent-like structures to withstand the draw of the vacuum. The collapse of the outshell negates any effort to reduce pressure on the body as the the crushing actually places the occupying limb into compression. Over nine protype combinations were evaluated including foams, spacer mech rigid framing material, and an inflatable design. To date, the rigid frame construction has been able to with stand 4 HP vaccum as well as 6 HP without crushing. Finalization of two prototypes will be prioritized and delivered to Dr. Levine's group for further validation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Dr. Levine's group is investigating Earth bound applications that could use the system designed in this project. Long term commercial uses could include hospital adoption, and long term recovery benefit for athletes may be researched by Under Armour.
Task Progress:	Major accomplishments: Over nine prototype combinations were evaluated including foams, spacer mesh, rigid framing material, and an inflatable design. To date the rigid frame construction has been able to withstand 4 HP vacuum as well as 6 HP without crushing. Finalization of two prototypes will be prioritized and delivered to Dr. Levine's group for further validation. Deliverables: Prototype with rigid support and inflatable version to be delivered under vacuum. Milestones: First prototype delivered June 2015 crushed under vacuum. Redesign pending for delivery. Customer Supplier Agreements: NA. Risk Reduction/Gap Closure: Risk reduction has not yet been identified.
Bibliography Type:	Description: (Last Updated:)